THE PICARD STACK IS REPRESENTABLE BY OPEN IMMERSIONS INTO THE COHERENT STACK

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Section examples-stacks-section-picard-stack (canonical tag 0372). cited at proof line 2830. The triples (T, g, L) of the Picard stack.
- Situation situation-coherent (canonical tag 08KB). cited at proof line 2833. The same triple viewed as an object of the coherent stack.
- Lemma spaces-morphisms-lemma-base-change-flat (canonical tag 03MO). cited at proof line 2837. X_T -> T flat makes L flat over T.
- Lemma spaces-morphisms-lemma-base-change-proper (canonical tag 04WP). cited at proof line 2838. X_T -> T proper makes the support of L proper over T.
- Lemma spaces-more-morphisms-lemma-finite-free-open (canonical tag 0CZT). cited at proof lines 2854-2855. The open W where F is finite locally free, with base-change compatibility.

Source reference. Original section 0372: sources/stacks-chow/tags.txt, source lines 4185–4185. Cited at proof line 2830: the triples (T, g, L) defining the Picard stack; load-bearing.

Source reference. Original definition 08KB: sources/stacks-chow/tags.txt, source lines 10881–10881. Cited at proof line 2833: the triple is viewed as an object of the coherent stack situation; load-bearing.

Source reference. Original lemma 03MO: sources/stacks-chow/tags.txt, source lines 4756–4756. Cited at proof line 2837: L is flat over T because X_T -> T is flat; load-bearing.

Source reference. Original lemma 04WP: sources/stacks-chow/tags.txt, source lines 6367–6367. Cited at proof line 2838: the support of L is proper over T; load-bearing.

Source reference. Original lemma 0CZT: sources/stacks-chow/tags.txt, source lines 16288–16288. Cited at proof lines 2854-2855: W is open and its formation commutes with base change; load-bearing.

Date: stacks-project:quot.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let S be a scheme. Let $f : X \to B$ be a morphism of algebraic spaces over S which is flat, of finite presentation, and proper. The natural map*

$$\mathcal{P}ic_{X/B} \longrightarrow \mathcal{C}oh_{X/B}$$

is representable by open immersions.

Proof. Observe that the map simply sends a triple $(T, g, \mathcal{L})$ as in Examples of Stacks, Section 0372 to the same triple $(T, g, \mathcal{L})$ but where now we view this as a triple of the kind described in Situation 08KB. This works because the invertible $\mathcal{O}_{X_T}$-module $\mathcal{L}$ is certainly a finitely presented $\mathcal{O}_{X_T}$-module, it is flat over T because $X_T \to T$ is flat, and the support is proper over T as $X_T \to T$ is proper (Morphisms of Spaces, Lemmas 03MO and 04WP). Thus the statement makes sense.

Having said this, it is clear that the content of the lemma is the following: given an object $(T, g, \mathcal{F})$ of $\mathcal{C}oh_{X/B}$ there is an open subscheme $U \subset T$ such that for a morphism of schemes $T' \to T$ the following are equivalent

(a) $T' \to T$ factors through U,
(b) the pullback $\mathcal{F}_{T'}$ of $\mathcal{F}$ by $X_{T'} \to X_T$ is invertible.

Let $W \subset |X_T|$ be the set of points $x \in |X_T|$ such that $\mathcal{F}$ is locally free in a neighbourhood of x. By More on Morphisms of Spaces, Lemma 0CZT. W is open and formation of W commutes with arbitrary base change. Clearly, if $T' \to T$ satisfies (b), then $|X_{T'}| \to |X_T|$ maps into W. Hence we may replace T by the open $T \setminus f_T(|X_T| \setminus W)$ in order to construct U. After doing so we reach the situation where $\mathcal{F}$ is finite locally free. In this case we get a disjoint union decomposition $X_T = X_0 \amalg X_1 \amalg X_2 \amalg \ldots$ into open and closed subspaces such that the restriction of $\mathcal{F}$ is locally free of rank i on X_i. Then clearly

$$U = T \setminus f_T(|X_0| \cup |X_2| \cup |X_3| \cup \ldots)$$

works. (Note that if we assume that T is quasi-compact, then X_T is quasi-compact hence only a finite number of X_i are nonempty and so U is indeed open.) $\square$